

1. A 2-year-old girl is brought to the office by her mother for a follow-up examination. Five days ago, she completed a 10-day course of amoxicillin for acute otitis media of the right ear. She has had three episodes of otitis media during the past year. She is not allergic to any medications. Immunizations are up-to-date. Growth and development are appropriate for age. On examination, the tympanic membranes are pearly gray and retracted; there is decreased mobility bilaterally. Which of the following is the most appropriate next step in management?
- ☐ A) Adenoidectomy
  - ☐ B) Amoxicillin therapy
  - ☐ C) Cefdinir therapy
  - ☐ D) Prednisone therapy
  - ☐ E) No management is indicated at this time



Next

<https://t.me/USMLENBME2CK>



Lab Values



Calculator



Review



Help



Pause



2. A 6-year-old girl is brought to the office by her mother because of compulsive eating since the age of 3 years. She was born following an uncomplicated pregnancy; delivery was complicated by breech presentation. During infancy, the patient had hypotonia. She began walking at the age of 2 years and talking at the age of 2½ years. The patient lives with her mother and older sister. She is at the 52nd percentile for height and 98th percentile for weight. Her temperature is 37°C (98.6°F), pulse is 92/min, respirations are 20/min, and blood pressure is 86/50 mm Hg. Physical examination shows almond-shaped eyes and a prominent nasal bridge. Examination of the extremities shows hypotonia. Mental status examination shows delayed speech and motor development. Which of the following is the most likely diagnosis?

- ☐ A) Angelman syndrome
- ☐ B) Down syndrome
- ☐ C) Fetal alcohol syndrome
- ☐ D) Prader-Willi syndrome
- ☐ E) Rett disorder
- ☐ F) Williams syndrome

3. Five minutes after delivery, a full-term newborn shows peripheral cyanosis, decreased muscle tone, and decreased reflex activity. His pulse is 80/min, and respirations are irregular. Which of the following is the most likely Apgar score?

- ☐ A) 0–1
- ☐ B) 2–3
- ☐ C) 4–6
- ☐ D) 7–8
- ☐ E) 9–10

4. An asymptomatic 14-year-old boy is brought for a well-child examination. He has had a murmur since 4 years of age. A grade 2/6, aortic systolic ejection murmur and a grade 1/6, early diastolic murmur at the midleft sternal border are heard. Echocardiography shows a bicuspid aortic valve with mild gradient. For which of the following conditions is this patient at greatest risk during childhood?
- ☐ A) Bacterial endocarditis
  - ☐ B) Brain abscess
  - ☐ C) Congestive heart failure
  - ☐ D) Dissecting aortic aneurysm
  - ☐ E) Syncope



5. A 2-year-old boy is brought to the physician because of a 2-day history of fever and mouth sores. His mother says he has been fussy and refused to eat but that he is beginning to feel better. He has a 1-year history of similar episodes that occur every 3 weeks and last 3 to 5 days. Growth and development are appropriate for age. His temperature is 38.4°C (101.2°F) , pulse is 120/min, and respirations are 20/min. Examination shows scattered aphthous ulcers in the oral cavity. The skin is clear, and there is no lymphadenopathy. Laboratory studies show:

|                       |                         |
|-----------------------|-------------------------|
| Hemoglobin            | 11.5 g/dL               |
| Leukocyte count       | 4000/mm <sup>3</sup>    |
| Segmented neutrophils | 4%                      |
| Lymphocytes           | 84%                     |
| Monocytes             | 12%                     |
| Platelet count        | 250,000/mm <sup>3</sup> |

Which of the following is the most likely diagnosis?

- ☐ A) Chédiak-Higashi syndrome
- ☐ B) Cyclic neutropenia
- ☐ C) Dyskeratosis congenita
- ☐ D) Fanconi syndrome
- ☐ E) Hypersplenism
- ☐ F) Leukemia
- ☐ G) Severe congenital neutropenia
- ☐ H) Viral illness
- ☐ I ) Vitamin B<sub>12</sub> (cobalamin) deficiency

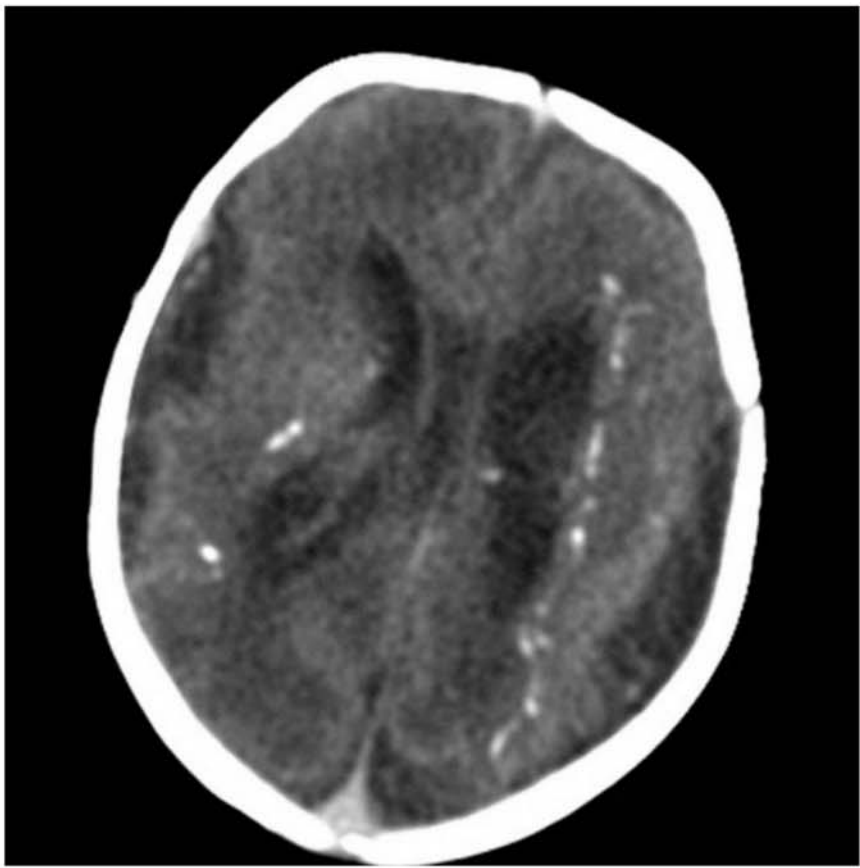


6. A previously healthy 6-year-old girl is brought to the physician in June because of a 4-day history of persistent fever and severe headache and a 1-day history of rash over her hands. She lives in a rural area and is frequently bitten by mosquitoes. One month ago, she was treated with corticosteroids for poison ivy. She appears ill. Her temperature is 39°C (102.2°F), pulse is 110/min, and respirations are 22/min. Examination shows bilateral, nonpurulent conjunctival injections and scattered petechial lesions over the palms and distal aspect of the upper extremities. Which of the following is the most likely diagnosis?
- ☐ A) Dengue fever
  - ☐ B) Erysipelas
  - ☐ C) Erythema infectiosum (fifth disease)
  - ☐ D) Exanthema subitum
  - ☐ E) Measles
  - ☐ F) Meningococcemia
  - ☐ G) Mucocutaneous lymph node syndrome (Kawasaki disease)
  - ☐ H) Rocky Mountain spotted fever
  - ☐ I ) Typhoid fever

7. A 4-year-old boy is brought to the office for a well-child visit. He can imitate a straight line, but he cannot copy a circle. He combines words, and only half his speech can be understood. He can jump but cannot balance on one foot. Physical examination shows no abnormalities. Which of the following is the most accurate assessment of this patient's development?

|                          | Gross<br>Motor Development | Fine<br>Motor Development | Language<br>Development |
|--------------------------|----------------------------|---------------------------|-------------------------|
| <input type="radio"/> A) | Delayed                    | delayed                   | delayed                 |
| <input type="radio"/> B) | Delayed                    | delayed                   | normal                  |
| <input type="radio"/> C) | Delayed                    | normal                    | delayed                 |
| <input type="radio"/> D) | Delayed                    | normal                    | normal                  |
| <input type="radio"/> E) | Normal                     | delayed                   | delayed                 |
| <input type="radio"/> F) | Normal                     | delayed                   | normal                  |
| <input type="radio"/> G) | Normal                     | normal                    | delayed                 |
| <input type="radio"/> H) | Normal                     | normal                    | normal                  |









8. A female newborn is admitted to the neonatal intensive care unit because of diffuse jaundice and petechial rash noted at birth. She was born at 35 weeks' gestation following spontaneous vaginal delivery to a 23-year-old woman, gravida 1, para 1. Pregnancy was complicated by fetal growth restriction. There is no family history of serious illness. There are no pets in the home. The patient is below the 3rd percentile for length, weight, and head circumference. Vital signs are within normal limits. Examination shows diffuse jaundice and a petechial rash over the head, trunk, and all extremities. No murmurs are heard. The liver edge is palpated 5 cm below the right costal margin. The spleen tip is palpated 3 cm below the left costal margin. Laboratory studies show:

|                  |                                      |
|------------------|--------------------------------------|
| Hemoglobin       | 16 g/dL (N=15–24)                    |
| Leukocyte count  | 8300/mm <sup>3</sup> (N=9100–34,000) |
| Platelet count   | 78,000/mm <sup>3</sup>               |
| Serum            |                                      |
| Bilirubin, total | 16 mg/dL                             |
| Direct           | 7.3 mg/dL                            |

A CT scan of the head is shown. Brain stem auditory evoked potential testing shows sensorineural hearing loss bilaterally. Congenital infection with which of the following is the most likely cause of this patient's condition?

- ☐ A) Cytomegalovirus
- ☐ B) *Listeria monocytogenes*
- ☐ C) Parvovirus B19
- ☐ D) Rubella
- ☐ E) *Streptococcus agalactiae* (group B)
- ☐ F) *S. intermedius*
- ☐ G) *Toxoplasma gondii*



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

9. A 3-month-old boy is brought to the emergency department because of a 3-day history of pale complexion, low-grade fever, vomiting, occasional diarrhea, and poor feeding. On arrival, he appears mildly dehydrated and cyanotic. His temperature is 37.8°C (100°F), pulse is 165/min, respirations are 35/min, and blood pressure is 80/48 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 85%. A grade 3/6, harsh systolic ejection murmur is heard over the entire precordium. Perfusion and muscle tone are normal. During attempts to insert an intravenous catheter, the patient begins to cry and becomes agitated. His oxygen saturation decreases to 60%, and the murmur becomes barely audible. Supplemental oxygen is administered. Which of the following is the most likely diagnosis?
- ☐ A) Atrial septal defect
  - ☐ B) Coarctation of the aorta
  - ☐ C) Pulmonary stenosis
  - ☐ D) Tetralogy of Fallot
  - ☐ E) Ventricular septal defect



10. A 5-day-old boy of Mediterranean descent is brought to the physician because of a 2-day history of progressive jaundice. He has had five yellow, seedy stools daily during this period. He was born at term to a 32-year-old primigravid woman. During the third trimester, his mother took an old prescription of sulfamethoxazole for cystitis without consulting the physician. The blood group of the mother and the newborn is O, Rh-positive. He receives 2 ounces of a cow's milk-based formula every 2 hours. Examination shows no abnormalities except for jaundice from the head to the upper thighs. His hematocrit is 33%, and total serum bilirubin concentration is 20 mg/dL with a direct component of 0.9 mg/dL. Direct and indirect antiglobulin (Coombs) tests are negative. Which of the following is the most likely explanation for this patient's jaundice?
- ☐ A) Dehydration
  - ☐ B) Enzyme deficiency
  - ☐ C) Milk protein allergy
  - ☐ D) Physiologic jaundice
  - ☐ E) Sepsis



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

11. A 7-year-old boy is brought to the physician for a follow-up examination 3 weeks after a nail punctured his right foot through his shoe. The wound was thoroughly cleaned and bandaged, but he continues to have moderate pain at the site. The patient received the tetanus vaccine at age 5 years and is otherwise healthy. His temperature is 37.2°C (99°F); other vital signs are within normal limits. Examination of the wound site shows erythema and warmth of the overlying skin with tenderness to palpation. Which of the following is the most likely causal organism?
- ☐ A) *Aeromonas hydrophila*
  - ☐ B) Group A streptococcus
  - ☐ C) *Klebsiella pneumoniae*
  - ☐ D) *Pseudomonas aeruginosa*
  - ☐ E) *Staphylococcus aureus*

12. A 16-year-old girl is brought to the physician by her mother because she has not yet had a menstrual period. She has no history of serious illness and takes no medications. She appears in no distress. She is 170 cm (5 ft 7 in) tall and weighs 54 kg (120 lb); BMI is 19 kg/m<sup>2</sup>. Breast development is sexual maturity rating stage 4, and there is minimal axillary and pubic hair development. Pelvic examination shows normal external female genitalia. The cervix is not visible, and no pelvic masses are palpable. Which of the following is the most appropriate next step in management?
- ☐ A) Follow-up examination in 6 months
  - ☐ B) Karyotype analysis
  - ☐ C) Measurement of serum estrogen concentration
  - ☐ D) MRI of the pituitary gland
  - ☐ E) Estrogen therapy



13. A 14-year-old girl is brought to the physician because of dark-colored urine for 5 days and a facial rash for 1 day. She has a history of diffuse joint pain treated with acetaminophen. Her blood pressure is 115/70 mm Hg. She appears pale and thin. Examination shows a prominent, erythematous, butterfly-shaped macular eruption over the face. Which of the following laboratory findings is most likely to be decreased?

- ☐ A) Eosinophil count
- ☐ B) Factor D
- ☐ C) Factor P
- ☐ D) Serum C3 concentration
- ☐ E) Serum C9 concentration



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause



14. A 9-year-old boy is brought to the emergency department by paramedics 20 minutes after the sudden onset of severe shortness of breath after walking up a flight of steps. He has had similar episodes over the past 12 weeks. En route, administration of 40% oxygen by face mask did not relieve his symptoms. He appears anxious and restless. His temperature is 37°C (98.6°F), pulse is 110/min, respirations are 30/min, and blood pressure is 110/60 mm Hg. Faint expiratory wheezing is heard. Breath sounds are distant bilaterally. Which of the following is the most likely underlying mechanism of this patient's condition?
- ☐ A) Activation of alveolar macrophages
  - ☐ B) Activation of mast cells
  - ☐ C) Increased neutrophil-endothelial cell adhesion
  - ☐ D) Increased pulmonary capillary hydrostatic pressure
  - ☐ E) Increased pulmonary vascular resistance
  - ☐ F) Parenchymal infiltration by lymphocytes



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause



15. An 8-day-old newborn is brought to the emergency department 10 minutes after she developed weakness and stopped breathing while being bottle-fed. On arrival, she is intubated and stabilized. During stabilization, she has a 3-minute episode of bicycling movements of the lower extremities and deviation of the eyes to the left. Phenobarbital is administered. Her pulse is 160/min, respirations are 42/min, and blood pressure is 60/40 mm Hg. Examination shows poor response to stimulation and decreased muscle tone. Deep tendon reflexes are brisk. Babinski sign is present bilaterally. A lumbar puncture is performed. Cerebrospinal fluid analysis shows:

|                       |                     |
|-----------------------|---------------------|
| Glucose               | 55 mg/dL            |
| Protein               | 65 mg/dL            |
| Erythrocyte count     | 350/mm <sup>3</sup> |
| Leukocyte count       | 70/mm <sup>3</sup>  |
| Segmented neutrophils | 20%                 |
| Lymphocytes           | 80%                 |

Treatment with antibiotics is begun, and she is admitted to the intensive care unit. Which of the following is the most likely causal organism?

- ☐ A) Cytomegalovirus
- ☐ B) *Escherichia coli*
- ☐ C) Group B streptococcus
- ☐ D) Herpes simplex virus
- ☐ E) *Toxoplasma gondii*

16. A previously healthy 10-month-old boy is brought to the emergency department because of a 3-day history of profuse watery diarrhea, frequent vomiting, and poor feeding. He usually breast-feeds well but refused all feedings today. Growth and development are appropriate for age. He appears lethargic. His pulse is 180/min, respirations are 48/min, and blood pressure is 68/54 mm Hg. Examination shows mild abdominal distention. Capillary refill time is 4 seconds. Which of the following is the most appropriate next step in management?
- ☐ A) Oral rehydration
  - ☐ B) Continuous infusion of fluids at maintenance doses
  - ☐ C) Intravenous administration of dopamine
  - ☐ D) Intravenous administration of fresh frozen plasma
  - ☐ E) Intravenous bolus of 0.9% saline



17. An 8-year-old girl is brought to the physician because of a 6-day history of occasional headache, fatigue, and poor appetite and a 2-day history of brown urine. She vomited yesterday morning but has not had abdominal pain, diarrhea, or painful urination. Her father reports that she had a sore throat and fever 2 weeks ago that resolved spontaneously. She has no history of serious illness and takes no medications. Her growth and development have been appropriate for age. She appears mildly pale but is in no distress. She is at the 50th percentile for height and 75th percentile for weight. Her temperature is 36.7°C (98°F), pulse is 108/min, respirations are 16/min, and blood pressure is 138/88 mm Hg. The tonsils are enlarged and mildly erythematous, but no exudate is present. The lungs are clear to auscultation. On cardiac examination, a grade 2/6, systolic ejection murmur is heard; there is no rub or gallop. Abdominal examination shows no abnormalities. The extremities are warm; there is 1+ edema. Laboratory studies show:

|                               |                       |
|-------------------------------|-----------------------|
| Serum                         |                       |
| Na <sup>+</sup>               | 132 mEq/L             |
| K <sup>+</sup>                | 5.2 mEq/L             |
| Cl <sup>-</sup>               | 100 mEq/L             |
| HCO <sub>3</sub> <sup>-</sup> | 15 mEq/L              |
| Urea nitrogen                 | 42 mg/dL              |
| Creatinine                    | 1.2 mg/dL             |
| Urine                         |                       |
| pH                            | 5                     |
| Specific gravity              | 1.020                 |
| Blood                         | 2+                    |
| Protein                       | 1+                    |
| RBC casts                     | too numerous to count |
| Gram stain                    | no organisms          |

Which of the following is the most appropriate next step in diagnosis?



|                               |                       |
|-------------------------------|-----------------------|
| Serum                         |                       |
| Na <sup>+</sup>               | 132 mEq/L             |
| K <sup>+</sup>                | 5.2 mEq/L             |
| Cl <sup>-</sup>               | 100 mEq/L             |
| HCO <sub>3</sub> <sup>-</sup> | 15 mEq/L              |
| Urea nitrogen                 | 42 mg/dL              |
| Creatinine                    | 1.2 mg/dL             |
| Urine                         |                       |
| pH                            | 5                     |
| Specific gravity              | 1.020                 |
| Blood                         | 2+                    |
| Protein                       | 1+                    |
| RBC casts                     | too numerous to count |
| Gram stain                    | no organisms          |

Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Measurement of serum concentration of C3
- ☐ B) Serum antinuclear antibody assay
- ☐ C) Urine culture
- ☐ D) Echocardiography
- ☐ E) Renal ultrasonography



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause



18. An 18-month-old boy is brought to the emergency department because of a 2-day history of increasingly rapid breathing and irritability. Two months ago, he had a high-grade fever, "pink eye," red swollen hands, enlarged lymph nodes, and cracked red lips; these symptoms resolved spontaneously within 14 days. He is in obvious discomfort. His temperature is 38°C (100.4°F), pulse is 150/min, respirations are 60/min, and blood pressure is 60/40 mm Hg. Examination shows no other abnormalities. An x-ray of the chest shows cardiomegaly. An ECG shows right-axis deviation, left ventricular hypertrophy, and ST-segment depression in leads II and III. Which of the following is the most likely mechanism of this patient's tachypnea?
- ☐ A) Chronic inflammatory interstitial lung disease
  - ☐ B) Congestive heart failure
  - ☐ C) Mitral valve obstruction
  - ☐ D) Pericardial effusion
  - ☐ E) Thickened internal layer of pulmonary arterioles



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

19. A 12-year-old boy is brought to the office because of a 2-day history of severe bilateral calf pain. One week ago, he had a dry cough, sore throat, headache, and fever, which resolved with ibuprofen therapy after 5 days. Currently, he is not taking any medications. His temperature is 37°C (98.6°F). Examination shows marked tenderness to palpation of the calves. Which of the following is the most likely diagnosis?
- ☐ A) Dermatomyositis
  - ☐ B) Influenza-induced myositis
  - ☐ C) Mucocutaneous lymph node syndrome (Kawasaki disease)
  - ☐ D) Parvovirus B19-induced arthritis
  - ☐ E) Rheumatic fever



20. A 12-month-old girl is scheduled for discharge 7 days after admission to the hospital for treatment of bacterial pneumonia. She has responded well to treatment with intravenous antibiotics. This is her third admission to the hospital for treatment of bacterial pneumonia during the past 9 months. She has received no immunizations. Examination shows no abnormalities. Serum studies show a decreased T-lymphocyte function. Additional serum studies show:

|                         |                              |
|-------------------------|------------------------------|
| IgA                     | 50 mg/dL (N=11–106)          |
| IgG                     | 200 mg/dL (N=172–1069)       |
| IgM                     | 50 mg/dL (N=41–173)          |
| CD4+ T-lymphocyte count | 100/mm <sup>3</sup> (N≥1000) |

Which of the following vaccines should be administered at this time?

- ☐ A) BCG
- ☐ B) *Haemophilus influenzae* type b
- ☐ C) Rotavirus
- ☐ D) Varicella
- ☐ E) No vaccines can be safely administered



21. A 5-year-old girl is brought to the clinic by her mother because of a 5-day history of progressive burning with urination and redness in the vaginal area. Her mother also noticed a small amount of yellow discharge on the patient's underwear. The patient has not had fever, vomiting, or diarrhea. She has no history of serious illness and receives no medications. There is no history of sexual abuse. Vital signs are within normal limits. Genitourinary examination shows moderate erythema over the vagina and entire perineum, including the perianal area. No vesicles or pustules are noted. There is a small amount of white discharge at the introitus. The remainder of the examination shows no abnormalities. Results of a culture of the vaginal discharge are pending. Which of the following is the most appropriate pharmacotherapy?

- ☐ A) Intramuscular ceftriaxone and oral azithromycin
- ☐ B) Oral acyclovir
- ☐ C) Oral amoxicillin
- ☐ D) Oral metronidazole
- ☐ E) Topical nystatin



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause



22. A previously healthy 7-year-old boy is brought to the physician by his mother because of a 5-day history of runny nose and mild cough. His mother says his nasal discharge was clear at first but has been yellow for 24 hours. He had a low-grade fever for the first 2 days of his illness but has not had fever for 3 days. The patient's father recently had a cold. The patient is at the 50th percentile for height and weight. His temperature is 36.9°C (98.4°F), pulse is 100/min, respirations are 24/min, and blood pressure is 96/62 mm Hg. Examination shows a moderate amount of yellow-to-white discharge from both nostrils. There is no periorbital edema or erythema. Multiple 1.5-cm, freely mobile, nontender nodules are palpated in the anterior cervical area bilaterally. The remainder of the examination shows no abnormalities. Which of the following is the most likely diagnosis?

- ☐ A) Allergic rhinitis
- ☐ B) Allergic sinusitis
- ☐ C) Bacterial sinusitis
- ☐ D) Cervical lymphadenitis
- ☐ E) Viral rhinosinusitis



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause



23. An asymptomatic 17-year-old boy comes to the physician for a routine examination prior to participation in school sports. Examination shows no abnormalities except for approximately 10 light brown hyperpigmented lesions over the trunk and extremities; the lesions vary in size from 0.6 to 2 cm in diameter. There are also some freckle-like lesions over both axillae. Compared to the general population, this patient is at increased risk for which of the following?

- ☐ A) Arteriovenous malformation
- ☐ B) Cerebral aneurysm
- ☐ C) Craniopharyngioma
- ☐ D) Medulloblastoma
- ☐ E) Optic nerve glioma



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause



24. A 14-year-old boy is brought to the physician because of a 1-week history of a mildly pruritic rash on his left arm. He has mild asthma that is well controlled with an inhaled corticosteroid. He plays several sports and currently participates on the school wrestling team. His family lives on a farm in North Dakota. They have had a pet dog for 5 years; the dog is in good health and has no skin lesions. The patient appears well. His temperature is 37°C (98.6°F). Examination shows a 3-cm lesion over the left forearm. A photograph of the lesion is shown. Which of the following is the most appropriate pharmacotherapy?

- ☐ A) Topical clotrimazole
- ☐ B) Topical mupirocin
- ☐ C) Topical nystatin
- ☐ D) Oral cephalexin
- ☐ E) Oral itraconazole



25. A 3-year-old girl is brought to the office for a follow-up examination. She has a seizure disorder and strabismus. Her only medication is valproic acid. She is at the 25th percentile for height and weight and 3rd percentile for head circumference. She appears happy, has an excitable demeanor, and frequently smiles and laughs. Examination shows a flattened posterior aspect of the head, prominent jaw, and widely spaced teeth. She flaps her hands repeatedly. She has a stiff-legged gait and cannot climb stairs without assistance. Her attention span is short. She uses some gestures and sounds to communicate but has not spoken any words. Which of the following is the most likely genetic defect in this patient?

- ☐ A) Chromosome 7 deletion
- ☐ B) Chromosome 15 deletion, maternally inherited
- ☐ C) Chromosome 15 deletion, paternally inherited
- ☐ D) Mutation of *MECP2* gene
- ☐ E) Trisomy 15
- ☐ F) Trisomy 21



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause



26. A 2-year-old white girl is admitted to the hospital for treatment of pneumococcal bacteremia. Over the past year, she has had three episodes of acute otitis media, two episodes of bacterial pneumonia, and recurrent episodes of thrush. Her mother has a history of persistent oral candidiasis and chronic diarrhea and has had a 9-kg (20-lb) weight loss over the past 3 months. The patient is at the 50th percentile for height and 5th percentile for weight. The oral mucous membranes are covered with white plaques. Examination shows 2-cm, nontender, axillary lymph nodes. The spleen is palpated 2 cm below the left costal margin, and the liver is palpated 3 cm below the right costal margin. Laboratory studies show:

|                       |                         |
|-----------------------|-------------------------|
| Leukocyte count       | 5000/mm <sup>3</sup>    |
| Segmented neutrophils | 65%                     |
| Lymphocytes           | 35%                     |
| Platelet count        | 95,000/mm <sup>3</sup>  |
| Serum quantitative    |                         |
| Hemoglobin            |                         |
| IgA                   | 420 mg/dL (N=14-159)    |
| IgG                   | 2200 mg/dL (N=345-1236) |
| IgM                   | 400 mg/dL (N=43-207)    |

Which of the following mechanisms of disease is the most likely cause of this patient's symptoms?

- ☐ A) Chédiak-Higashi syndrome
- ☐ B) Chronic granulomatous disease
- ☐ C) Common variable immunodeficiency
- ☐ D) Cystic fibrosis
- ☐ E) HIV infection

margin, and the liver is palpated 2 cm below the right costal margin. Laboratory studies show:

|                       |                         |
|-----------------------|-------------------------|
| Leukocyte count       | 5000/mm <sup>3</sup>    |
| Segmented neutrophils | 65%                     |
| Lymphocytes           | 35%                     |
| Platelet count        | 95,000/mm <sup>3</sup>  |
| Serum quantitative    |                         |
| Hemoglobin            |                         |
| IgA                   | 420 mg/dL (N=14-159)    |
| IgG                   | 2200 mg/dL (N=345-1236) |
| IgM                   | 400 mg/dL (N=43-207)    |

Which of the following mechanisms of disease is the most likely cause of this patient's symptoms?

- ☐ A) Chédiak-Higashi syndrome
- ☐ B) Chronic granulomatous disease
- ☐ C) Common variable immunodeficiency
- ☐ D) Cystic fibrosis
- ☐ E) HIV infection
- ☐ F) Severe combined immunodeficiency disease
- ☐ G) Terminal complement deficiency
- ☐ H) Wiskott-Aldrich syndrome

27. A 2-month-old boy is brought to the physician for an initial examination because of persistent jaundice. He is described as a "good" baby, who breast-feeds reasonably well but often needs to be awakened to feed. Examination shows an umbilical hernia, mild jaundice, and cool extremities. Growth is appropriate for age. Which of the following is the most appropriate next step in diagnosis?
- ☐ A) Karyotype analysis
  - ☐ B) Measurement of sweat chloride concentration
  - ☐ C) Thyroid function tests
  - ☐ D) CT scan of the head
  - ☐ E) CT scan of the liver



28. A 3-week-old girl is admitted to the hospital because of a 1-day history of fever and irritability and a 2-day history of diarrhea that has become bloody during the past 6 hours. She was born at 39 weeks' gestation; pregnancy and delivery were uncomplicated. Her diet consists of cow milk-based formula. The family occasionally eats out at fast-food restaurants. There are two pet lizards at home. The patient is awake but irritable. She is at the 25th percentile for length and weight and 50th percentile for head circumference. Her temperature is 39.1°C (102.4°F), pulse is 170/min, and respirations are 40/min. Capillary refill time is less than 2 seconds. Her hemoglobin concentration is 16 g/dL, leukocyte count is 18,000/mm<sup>3</sup>, and platelet count is 260,000/mm<sup>3</sup>. Serum concentrations of electrolytes, urea nitrogen, and creatinine are within the reference ranges. After 24 hours, blood and stool cultures grow gram-negative rods. Which of the following is the most likely causal organism?

- ☐ A) *Campylobacter jejuni*
- ☐ B) *Escherichia coli* O157:H7
- ☐ C) *Listeria monocytogenes*
- ☐ D) *Salmonella enteritidis*
- ☐ E) *Shigella sonnei*



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause



29. A 2-day-old full-term newborn is brought to the physician because of bilateral eye discharge. He was delivered at home by his father and has not yet been examined by a physician. He appears well. Vital signs are within normal limits. Examination shows marked swelling of the eyelids; there is profuse yellow discharge from both eyes. The conjunctivae cannot be visualized because of the swelling. No other abnormalities are noted. Which of the following is the most likely diagnosis?

- ☐ A) Allergic conjunctivitis
- ☐ B) Chemical conjunctivitis
- ☐ C) Chlamydial conjunctivitis
- ☐ D) Gonococcal conjunctivitis
- ☐ E) Mucocutaneous lymph node syndrome (Kawasaki disease)



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause





30. A 22-month-old boy has had fever, vomiting, and diarrhea for 3 days. His mother has been supplementing his usual diet with extra nondairy liquids during this period. He has had 12 to 15 diarrheal stools each day. She says that it is difficult to distinguish stool from urine when she changes his diaper. His temperature is 37.9°C (100.2°F), pulse is 120/min, and respirations are 32/min. Laboratory studies show:

|                               |           |
|-------------------------------|-----------|
| Serum                         |           |
| Na <sup>+</sup>               | 135 mEq/L |
| Cl <sup>-</sup>               | 115 mEq/L |
| K <sup>+</sup>                | 4.8 mEq/L |
| HCO <sub>3</sub> <sup>-</sup> | 10 mEq/L  |
| Urine pH                      | 5.0       |

Arterial blood gas analysis on room air shows:

|                               |           |
|-------------------------------|-----------|
| pH                            | 7.28      |
| Pco <sub>2</sub>              | 30 mm Hg  |
| Po <sub>2</sub>               | 108 mm Hg |
| HCO <sub>3</sub> <sup>-</sup> | 10 mEq/L  |

Which of the following is the most likely explanation for his acid-base disorder?

- ☐ A) Laboratory error
- ☐ B) Renal losses of bicarbonate
- ☐ C) Respiratory insufficiency
- ☐ D) Stool losses of bicarbonate
- ☐ E) Vomiting



31. A 5-day-old newborn has hemorrhagic disease due to vitamin K deficiency. He is bleeding from the nose, mouth, and gastrointestinal tract and into the subdural space. In addition to vitamin K therapy, administration of which of the following is the most appropriate next step?

- ☐ A) Cryoprecipitates
- ☐ B) Factor VIII concentrate
- ☐ C) Fresh frozen plasma
- ☐ D) Lactated Ringer solution
- ☐ E) Platelets

32. A 5-week-old boy is brought to the emergency department because of a 1-day history of vomiting immediately after each feeding. During this time, he has had only two wet diapers. He was born at term following an uncomplicated pregnancy and delivery and weighed 3317 g (7 lb 5 oz). He was exclusively breast-fed during his first 3 weeks of life; for the past 2 weeks, his diet has consisted of a cow milk-based formula. He has been taking 4 to 5 oz per feeding and, until yesterday, had minimal spitting up after each feeding. At his most recent examination 1 week ago, he weighed 4224 g (9 lb 5 oz). Today, he weighs 85 g (3 oz) less. Physical examination shows no other abnormalities. After the examination, he is given a bottle of formula, drinks 4 oz, and vomits a large amount. This patient most likely has an obstruction at which of the following locations?
- ☐ A) Duodenum
  - ☐ B) Gastroesophageal junction
  - ☐ C) Jejunum
  - ☐ D) Midesophagus
  - ☐ E) Pyloric junction



33. A 4-month-old female infant is brought to the physician because of a rash in the diaper area for 5 days. The infant has had two episodes of otitis media treated with oral antibiotic therapy but has been otherwise healthy. Her temperature is 37°C (98.6°F). Physical examination shows a red, confluent rash with sharply demarcated borders, pustules, and satellite papules in the diaper area. Which of the following is the most appropriate pharmacotherapy?
- ☐ A) Topical mupirocin
  - ☐ B) Topical nystatin
  - ☐ C) Oral cephalexin
  - ☐ D) Oral ketoconazole
  - ☐ E) Intravenous amphotericin B
  - ☐ F) No pharmacotherapy is indicated

34. A 2-year-old boy who is HIV positive is brought to the family clinic by his mother for a follow-up examination. He is currently taking three antiretroviral drugs. His HIV-positive mother is also taking multiple antiretroviral drugs; she has chronic cryptosporidiosis and has had a 5-kg (11-lb) weight loss during the past 2 months. The father is not in contact with the family. The child's growth and development are appropriate for age, and immunizations are up-to-date. Examination shows two small, nontender, mobile lymph nodes in the right axilla. Laboratory studies show:

|                            | Child                              | Mother                         |
|----------------------------|------------------------------------|--------------------------------|
| Leukocyte count            | 12,800/mm <sup>3</sup>             | 4100/mm <sup>3</sup>           |
| Segmented neutrophils      | 55%                                | 80%                            |
| Lymphocytes                | 35%                                | 10%                            |
| Monocytes                  | 10%                                | 10%                            |
| CD4+ T-lymphocyte count    | 1852/mm <sup>3</sup> (Normal≥1000) | 4/mm <sup>3</sup> (Normal≥500) |
| Plasma HIV viral load      | undetectable                       | 38,000 copies/mL               |
| Serum hepatitis C antibody | negative                           | positive                       |

Which of the following is the most appropriate topic to discuss at this visit?

- ☐ A) The advantages of three-drug versus four-drug antiretroviral regimens
- ☐ B) Care of the child in case of the mother's death
- ☐ C) The child's immunization schedule
- ☐ D) Hand-washing techniques to prevent HIV transmission
- ☐ E) Long-term complications of HIV infection



35. A 2-day-old female newborn is examined before being discharged home. She was born at term following an uncomplicated pregnancy and spontaneous vaginal delivery. She is 50 cm (20 in) in length and weighs 3515 g (7 lb 12 oz). Her temperature is 36.8°C (98.2°F), pulse is 128/min, respirations are 38/min, and blood pressure is 70/54 mm Hg. Examination shows a generalized rash with small yellow papules on an erythematous base, predominantly over the trunk. Which of the following is the most appropriate next step in management?

- ☐ A) Culture of lesions and antibiotic therapy to eradicate *Staphylococcus aureus*
- ☐ B) Culture of lesions for herpes simplex virus
- ☐ C) KOH stain of a lesion scraping for *Candida* species
- ☐ D) Topical corticosteroid therapy
- ☐ E) Reassurance to the parents that the rash will resolve in a few days



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause



36. A 5-week-old boy is brought to the emergency department by his parents in the fall because of a 2-week history of progressive cough. He often coughs uncontrollably until he is struggling to breathe and vomits. He has not had fever. He had moderate nasal congestion for the first 5 days of his illness. He was born following an uncomplicated pregnancy and delivery. The patient's father also has had a cough for 3 weeks. During the examination, the patient is not in distress except when he is coughing. He is at the 25th percentile for length, weight, and head circumference. His temperature is 37.1°C (98.8°F), pulse is 148/min, and respirations are 40/min. Pulse oximetry on room air shows an oxygen saturation of 95%. The lungs are clear to auscultation. The remainder of the examination shows no abnormalities. Laboratory studies show:

|                       |                                        |
|-----------------------|----------------------------------------|
| Leukocyte count       | 28,000/mm <sup>3</sup> (N=6000-14,000) |
| Segmented neutrophils | 14%                                    |
| Bands                 | 1%                                     |
| Lymphocytes           | 80%                                    |
| Monocytes             | 5%                                     |

A chest x-ray shows no abnormalities. Which of the following is the most likely causal organism?

- ☐ A) *Bordetella pertussis*
- ☐ B) Influenza virus
- ☐ C) *Mycoplasma pneumoniae*
- ☐ D) Respiratory syncytial virus
- ☐ E) *Streptococcus pneumoniae*





37. A 17-year-old boy comes to the office for an examination prior to participating in football. The patient says he feels well. He has no history of serious illness and takes no medications. There is no family history of serious illness. The patient is 180 cm (5 ft 11 in) tall and weighs 93 kg (205 lb); BMI is 29 kg/m<sup>2</sup>. His blood pressure in the right upper extremity, obtained by automated oscillometric device, is 138/88 mm Hg (95th percentile for age and height). Repeat blood pressure by the device is 140/89 mm Hg. Examination shows no other abnormalities. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Blood pressure measurement by auscultation
- ☐ B) Echocardiography
- ☐ C) 24-Hour ambulatory blood pressure monitoring
- ☐ D) Repeat blood pressure measurement by oscillometric device in the office daily for the next 3 days
- ☐ E) Repeat blood pressure measurement by oscillometric device in the office in 2 weeks



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

38. A 3-year-old boy is brought to the emergency department 4 hours after the onset of irritability and temperatures to 40°C (104°F). He received the diagnosis of systemic-onset juvenile idiopathic arthritis 2 months ago; treatment with ibuprofen has begun to improve his joint stiffness and swelling. He appears pale and is inconsolable. Examination shows scattered petechiae over the trunk and extremities, a distended abdomen, and diffuse lymphadenopathy. The liver edge and spleen tip are palpated 5 cm below the costal margins. There is bilateral knee and ankle swelling with effusions. Laboratory studies show:

|                                |                        |
|--------------------------------|------------------------|
| Hematocrit                     | 23.4%                  |
| Leukocyte count                | 2500/mm <sup>3</sup>   |
| Segmented neutrophils          | 20%                    |
| Bands                          | 10%                    |
| Lymphocytes                    | 70%                    |
| Platelet count                 | 45,000/mm <sup>3</sup> |
| Erythrocyte sedimentation rate | 12 mm/h                |
| Serum                          |                        |
| AST                            | 235 U/L                |
| ALT                            | 337 U/L                |
| C-reactive protein             | 3.5 mg/L (N=0.08–3.1)  |

Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Abdominal CT scan
- ☐ B) Bone scan
- ☐ C) Gallium scan
- ☐ D) Bone marrow aspiration
- ☐ E) Liver biopsy



39. A 7-year-old boy is brought to the emergency department because of the acute onset of pain over the anterior surface of the neck; the pain is exacerbated by swallowing or sticking out his tongue. He also has a 1-day history of fever and chills. Examination shows tenderness in the midline over the trachea in the region of the thyroid isthmus. There is cervical lymphadenopathy. He has scratches on his arm from playing with his 4-year-old cat. Which of the following is the most likely cause of these findings?

- ☐ A) Cat-scratch disease
- ☐ B) Infected thyroglossal duct cyst
- ☐ C) Infectious mononucleosis
- ☐ D) Tonsillitis
- ☐ E) Viral syndrome



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

40. A 6-month-old girl with Beckwith-Wiedemann syndrome is brought to the physician because of a 1-week history of intermittent vomiting and fussiness. She receives no medications. Vital signs are within normal limits. Examination shows macroglossia and right-sided hypertrophy. On abdominal examination, a firm mass is palpated in the right upper quadrant. Laboratory studies show:

|                                  |                      |
|----------------------------------|----------------------|
| $\beta$ -hCG                     | 0 mIU/mL             |
| $\alpha$ -Fetoprotein            | 15,678 ng/mL (N<10)  |
| Urine                            |                      |
| Homovanillic acid/creatinine     | 10 mg/g Cr (N=0–42)  |
| Vanillylmandelic acid/creatinine | 5.6 mg/g Cr (N=0–27) |

Abdominal ultrasonography is ordered. Which of the following is the most likely explanation for this patient's mass?

- ☐ A) Germ cell tumor
- ☐ B) Hepatoblastoma
- ☐ C) Nephroblastoma (Wilms tumor)
- ☐ D) Neuroblastoma
- ☐ E) Spina bifida occulta

41. A 12-year-old girl with chronic sinusitis is brought to the physician because of fever and productive cough for 4 days. She had pneumococcal pneumonia 5 months ago that was treated with intravenous ceftriaxone followed by oral amoxicillin. She is at the 50th percentile for height and weight. Her temperature is 39°C (102.2°F), pulse is 105/min, and respirations are 30/min. Breath sounds are decreased on the lower right. Laboratory studies show:

|                         |                                   |
|-------------------------|-----------------------------------|
| Leukocyte count         | 21,000/mm <sup>3</sup>            |
| Segmented neutrophils   | 70%                               |
| Bands                   | 15%                               |
| Lymphocytes             | 15%                               |
| Platelet count          | 140,000/mm <sup>3</sup>           |
| CD3+ T-lymphocyte count | 1500/mm <sup>3</sup> (N=820–1900) |
| CD4+ T-lymphocyte count | 1000/mm <sup>3</sup> (N≥500)      |
| CD8+ T-lymphocyte count | 400/mm <sup>3</sup> (N=250–650)   |
| Serum                   |                                   |
| IgA                     | 20 mg/dL                          |
| IgG                     | 150 mg/dL                         |
| IgM                     | 30 mg/dL                          |

Which of the following is the most likely diagnosis?

- ☐ A) Common variable immunodeficiency
- ☐ B) Mucocutaneous lymph node syndrome (Kawasaki disease)
- ☐ C) Severe combined immunodeficiency
- ☐ D) Systemic lupus erythematosus



|                         |                                   |
|-------------------------|-----------------------------------|
| Leukocyte count         | 21,000/mm <sup>3</sup>            |
| Segmented neutrophils   | 70%                               |
| Bands                   | 15%                               |
| Lymphocytes             | 15%                               |
| Platelet count          | 140,000/mm <sup>3</sup>           |
| CD3+ T-lymphocyte count | 1500/mm <sup>3</sup> (N=820–1900) |
| CD4+ T-lymphocyte count | 1000/mm <sup>3</sup> (N≥500)      |
| CD8+ T-lymphocyte count | 400/mm <sup>3</sup> (N=250–650)   |
| Serum                   |                                   |
| IgA                     | 20 mg/dL                          |
| IgG                     | 150 mg/dL                         |
| IgM                     | 30 mg/dL                          |

Which of the following is the most likely diagnosis?

- ☐ A) Common variable immunodeficiency
- ☐ B) Mucocutaneous lymph node syndrome (Kawasaki disease)
- ☐ C) Severe combined immunodeficiency
- ☐ D) Systemic lupus erythematosus
- ☐ E) Transient agammaglobulinemia of infancy
- ☐ F) X-linked agammaglobulinemia





42. An 8-year-old girl is brought to the physician because her parents are concerned that she is different from their older child and prefers to be alone instead of socializing with peers or family. She carries a doll that she found at a yard sale wherever she goes but never plays with it. Whenever she is without the doll, she becomes distraught and has a tantrum. The parents once invited a girl from their daughter's class to their home, but the patient appeared disinterested and would not play with the other girl. The patient's teachers report that she is often off task and distracted in her second-grade class but that she answers direct questions. She does not participate in games at recess, and she tends to drift off by herself and wander about the playground. She is at the 50th percentile for height, 30th percentile for weight, and 90th percentile for head circumference. Physical examination shows no abnormalities. During the examination, the patient holds an undressed doll by its feet. She makes little eye contact. She occasionally repeats the physician's words instead of answering questions. Which of the following is the most likely diagnosis?
- ☐ A) Learning disorder
  - ☐ B) Mild intellectual developmental disorder
  - ☐ C) Oppositional defiant disorder
  - ☐ D) Parent-child relational problem
  - ☐ E) Pervasive developmental disorder



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

43. A 10-year-old girl is brought to the physician by her mother 3 hours after she refused to get out of bed this morning. Over the past 4 days, she has had fever, malaise, and severe pain in her elbows, wrists, and knees. The family recently lived in a refugee camp. She appears flushed and acutely ill. Her pulse is 110/min, respirations are 20/min, and blood pressure is 100/40 mm Hg. The knee and ankle joints are warm and swollen. A grade 4/6 holosystolic murmur is heard at the apex and radiates to the left axilla. Which of the following is the most likely diagnosis?

- ☐ A) Dengue fever
- ☐ B) Lyme disease
- ☐ C) Malaria
- ☐ D) Mucocutaneous lymph node syndrome (Kawasaki disease)
- ☐ E) Rheumatic fever
- ☐ F) Typhoid fever

44. A previously healthy 7-week-old girl is brought to the emergency department by her parents because of a 3-day history of progressive vomiting. Her symptoms began with spitting up after every feeding; now she has forceful vomiting. Her mother says that she has seemed hungry during this period. She has not had fever or diarrhea. She is alert and appears well hydrated. Her temperature is 37.4°C (99.3°F), pulse is 120/min, and respirations are 28/min. Examination shows no other abnormalities. Laboratory studies show:

|                               |                         |
|-------------------------------|-------------------------|
| Hemoglobin                    | 9 g/dL                  |
| Leukocyte count               | 8500/mm <sup>3</sup>    |
| Segmented neutrophils         | 50%                     |
| Lymphocytes                   | 45%                     |
| Monocytes                     | 5%                      |
| Platelet count                | 360,000/mm <sup>3</sup> |
| Serum                         |                         |
| Na <sup>+</sup>               | 138 mEq/L               |
| Cl <sup>-</sup>               | 90 mEq/L                |
| K <sup>+</sup>                | 3.6 mEq/L               |
| HCO <sub>3</sub> <sup>-</sup> | 36 mEq/L                |
| Urea nitrogen                 | 25 mg/dL                |
| Glucose                       | 70 mg/dL                |
| Creatinine                    | 0.3 mg/dL               |

Which of the following is the most likely diagnosis?

- ☐ A) Acute hyperammonemia
- ☐ B) Acute viral gastroenteritis



|                               |                         |
|-------------------------------|-------------------------|
| Segmented neutrophils         | 88%                     |
| Lymphocytes                   | 45%                     |
| Monocytes                     | 5%                      |
| Platelet count                | 360,000/mm <sup>3</sup> |
| Serum                         |                         |
| Na <sup>+</sup>               | 138 mEq/L               |
| Cl <sup>-</sup>               | 90 mEq/L                |
| K <sup>+</sup>                | 3.6 mEq/L               |
| HCO <sub>3</sub> <sup>-</sup> | 36 mEq/L                |
| Urea nitrogen                 | 25 mg/dL                |
| Glucose                       | 70 mg/dL                |
| Creatinine                    | 0.3 mg/dL               |

Which of the following is the most likely diagnosis?

- ☐ A) Acute hyperammonemia
- ☐ B) Acute viral gastroenteritis
- ☐ C) Gastroesophageal reflux
- ☐ D) Hemolytic-uremic syndrome
- ☐ E) Malrotation of the intestine
- ☐ F) Porphyria
- ☐ G) Pyloric stenosis



45. A 17-year-old girl comes to the office because of a 5-day history of a nontender lesion on her vagina. She is sexually active with three male partners, and they use condoms inconsistently. Her temperature is 37.2°C (99°F), pulse is 78/min, respirations are 18/min, and blood pressure is 106/68 mm Hg. Examination shows a 2 x 2-cm ulcerated lesion with raised borders over the right labium majus. Which of the following is the most likely causal organism?

- ☐ A) Gram-negative bacillus
- ☐ B) Gram-negative diplococcus
- ☐ C) Obligate intracellular bacterium
- ☐ D) Protozoan
- ☐ E) Spirochete
- ☐ F) Virus



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause



46. A 12-year-old boy is brought for a well-child examination. Since his last visit 1 year ago, he has had no illnesses. His pulse is 102/min, respirations are 21/min, and blood pressure is 102/60 mm Hg. Examination shows no abnormalities. Urinalysis shows 3+ protein. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Repeat urinalysis at bedtime and on awakening
- ☐ B) Intravenous pyelography
- ☐ C) Renal ultrasonography
- ☐ D) Trial of corticosteroid therapy
- ☐ E) Renal biopsy



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause





47. A healthy 16-month-old girl is brought to the physician because of a painless right-sided limp since she began to walk 4 months ago. She was delivered at term by cesarean section because of breech presentation. There is no family history of musculoskeletal disorders. Examination shows a left lower extremity that is shorter than the right; sensation, circulation, and muscle strength are normal. Abduction of the left hip is limited to 30 degrees. Examination of the right hip shows no abnormalities. Which of the following is the most likely underlying mechanism of these findings?

- ☐ A) Acute epiphyseal fracture
- ☐ B) Acute suppurative arthritis
- ☐ C) Avulsion of muscle origin
- ☐ D) Displacement of the capital femoral epiphysis
- ☐ E) Failure of osteoid to mineralize
- ☐ F) Poorly developed acetabulum



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause



48. A 16-year-old girl is brought to the office by her mother for a health maintenance examination. She has no history of serious illness and takes no medications. Immunizations are up-to-date. Menses occur at irregular 28- to 35-day intervals. Her last menstrual period was 3 weeks ago. She does not smoke cigarettes, drink alcohol, or use illicit drugs. She has been sexually active with one male partner for 1 year; they do not use condoms. Her mother was diagnosed with breast cancer at the age of 40 years. The patient is 163 cm (5 ft 4 in) tall and weighs 65 kg (143 lb); BMI is 25 kg/m<sup>2</sup>. Vital signs are within normal limits. Breast examination shows no masses or discharge. The external genitalia appear normal; there is no vaginal discharge. Breast and pubic hair development are sexual maturity rating stage 5. Which of the following is the most appropriate screening test to administer at this time?

- ☐ A) Mammography
- ☐ B) Measurement of fasting serum glucose concentration
- ☐ C) Measurement of hemoglobin concentration
- ☐ D) Nucleic acid amplification testing for *Chlamydia trachomatis* and *Neisseria gonorrhoeae*
- ☐ E) Pap smear



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

49. A 17-year-old girl comes to the office because she has had a burning sensation at the beginning of urination for the last 24 hours. She has no history of similar symptoms. Examination shows no abnormalities. Urinalysis shows marked pus and no bacteria. Which of the following is the most appropriate next step in management?

- ☐ A) Sitz baths
- ☐ B) Testing for sexually transmitted diseases
- ☐ C) Wet mount preparation with KOH
- ☐ D) X-ray screening for urinary anomaly
- ☐ E) No intervention

50. A 6-month-old boy is brought to the office for a well-child examination. He is exclusively fed cow milk-based formula. His mother reports that he takes his bottle vigorously, but he frequently spits up after eating. He has not had diarrhea or fever. Development is appropriate for age. He has dropped from the 50th percentile to the 35th percentile for weight since his last visit 2 months ago. Length and head circumference remain at the 50th percentile. Examination shows decreased subcutaneous fat. No other abnormalities are noted. Which of the following is the most likely diagnosis?
- ☐ A) Gastroesophageal reflux disease
  - ☐ B) Inflammatory bowel disease
  - ☐ C) Necrotizing enterocolitis
  - ☐ D) Pancreatitis
  - ☐ E) Pyloric stenosis